Physics 5C Midterm 1 Review Session

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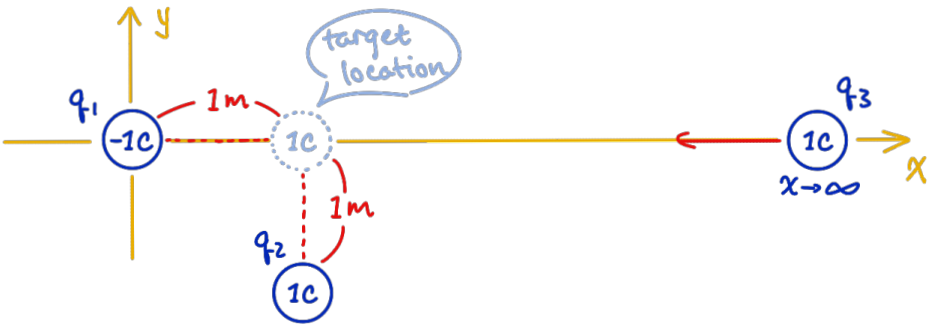
Consider the following configuration:
A charge $q_1 = -1\,\mathrm{C}$ is fixed at $\vec{x}_1 = (0, 0)\,\mathrm{m}$.
A charge $q_2 = 1\,\mathrm{C}$ is fixed at $\vec{x}_2 = (1, -1)\,\mathrm{m}$.
Jinyan is trying to bring in a test charge $q_3 = 1\,\mathrm{C}$ from infinitely far away at $\vec{x}_i = (\infty, 0)\,\mathrm{m}$ to the target location at $\vec{x}_f = (1, 0)\,\mathrm{m}$. We will discuss this process from two perspectives, first using electric potential and energy, then using electric field and forces.



First we consider the picture of electric potential.
Part (a) Compute the electric potential at $\vec{x}_i$ and $\vec{x}_f$ due to q_1 and q_2 combined.
Part (b) Compute the electric potential energy stored in q_3 at $\vec{x}_i$ and $\vec{x}_f$.
Part (c) How much energy is needed to bring q_3 from $\vec{x}_i$ to $\vec{x}_f$?
Part (d) If q_3 is fixed at $\vec{x}_f = (1, 0)\,\mathrm{m}$, compute the potential energy stored in q_2.
Part (e) What is the total energy needed to assemble the three-charge system? [1]

Now let us consider the picture of electric field.
Part (f) Draw the electric field lines due to the source charges q_1 and q_2 combined.
Part (g) Compute the net electric field due to q_1 and q_2 combined, at $\vec{x}_i$ and at $\vec{x}_f$.
Part (h) Compute the net force that q_3 experiences when it is at $\vec{x}_i$ and at $\vec{x}_f$.
Part (i) Jinyan flips the sign of q_3 when it is at $\vec{x}_f$. What gets changed in this picture?

Now we explore the relation between electric field and electric potential.
Consider only the source charges q_1 and q_2, that is, neglect the test charge q_3.
Part (j) Find the location (other than very far away) where we have the "net" electric potential being zero. Pick one of those locations and compute the magnitude of the electric field at that location. Qualitatively describe the relation between electric field and electric potential.

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[1] That is, this should be a three-step calculation: Step 1 - Starting with a vacuum, nothing in space, find the energy to put q_1 at the origin $\vec{x}_1$; Step 2 - With only q_1 fixed at the origin, find the energy needed to bring in q_2 from infinity to $\vec{x}_2 = (1, -1)\,\mathrm{m}$; Step 3 - With q_1 and q_2 fixed at their locations, find the energy needed to bring in q_3 from infinity to $\vec{x}_f$. Sum these energies to find the total energy stored in such a system.

Solution

Part (a) First we compute the electric potential due to the two source charges q_1 and q_2. Remember that electric potential is a scalar, so we just add the contributions from the two charges directly. For q_3 at infinity, it is infinitely far away from q_1 and q_2, so we can approximate the quantity $1/r$ to be zero, that is

$$V_i = \frac{kq_1}{r_{13}} + \frac{kq_2}{r_{23}} \approx \frac{kq_1}{\infty} + \frac{kq_2}{\infty} \approx 0\,\mathrm{V}\,.$$

At the final position $\vec{x}_f = (1, 0)\,\mathrm{m}$, the distance from q_1 to $\vec{x}_f$ is $1\,\mathrm{m}$, and the distance from q_2 to $\vec{x}_f$ is also $1\,\mathrm{m}$. Therefore,

$$V_f = \frac{kq_1}{1} + \frac{kq_2}{1} = \frac{k(-1\,\mathrm{C})}{1\,\mathrm{m}} + \frac{k(1\,\mathrm{C})}{1\,\mathrm{m}} = 0\,\mathrm{V}\,.$$

This cancellation happens because $\vec{x}_f$ is equally far away from the two charges, and the two charges have equal magnitude but opposite signs, but it does not mean that the charge q_3 feels a zero net force at $\vec{x}_f$, nor that the electric field at $\vec{x}_f$ is zero, as we will see in the second half of this problem.

Part (b) The electric potential energy of q_3 is

$$U = q_3 V\,.$$

Since $q_3 = 1\,\mathrm{C}$, we have

$$U_i = q_3 V_i = (1\,\mathrm{C})(0\,\mathrm{V}) = 0\,\mathrm{J}\,, \qquad U_f = q_3 V_f = (1\,\mathrm{C})(0\,\mathrm{V}) = 0\,\mathrm{J}\,.$$

Part (c) If we slowly bring q_3 from $\vec{x}_i$ to $\vec{x}_f$, the energy that we need to spend is the change in electric potential energy,

$$\text{External Work} = U_f - U_i = 0 - 0 = 0\,\mathrm{J}\,.$$

So in this particular case, it takes zero net energy to bring q_3 from infinity to $\vec{x}_f$. Again, this is a statement about energy difference, not a statement that the electric force is zero everywhere along the path. If you sketch the equipotential lines of the electric potential created by q_1 and q_2 combined, you will see that, when moving q_3 from infinity to $\vec{x}_f$, the electric potential energy stored in q_3 first increases (from 0 to some positive number E), then decreases (from the positive number E back to 0). This is similar to rolling a solid sphere up and down a mountain, but the starting point and the ending point have the same height, so the gravitational potential energy $U_g = mgh$ is the same at the two points, and thus it seems like it takes zero energy to move between the two points, but the full story is, initially you put in some work to move the sphere up to the top of the mountain, then gravity is helping you to move the sphere from the top of the mountain down to the final destination. In the first half of the journey, you perform positive work to counteract the gravity, and in the second half, you perform negative work which cancels exactly with the positive work done in the first half.

Part (d) Now suppose q_3 is fixed at $\vec{x}_f = (1, 0)\,\mathrm{m}$, and we ask about the potential energy stored in q_2. Then q_2 feels the electric potential due to the other two charges, q_1

and q_3. In this part, we are essentially treating q_2 as a test charge, with q_1 and q_3 being source charges. The distance between q_2 and q_1 is

$$r_{21} = \sqrt{(1-0)^2 + (-1-0)^2}\,\mathrm{m} = \sqrt{2}\,\mathrm{m}\,,$$

and the distance between q_2 and q_3 is

$$r_{23} = 1\,\mathrm{m}\,.$$

Therefore, the electric potential at the position of q_2 due to q_1 and q_3 is

$$V(\vec{x}_2) = \frac{kq_1}{r_{21}} + \frac{kq_3}{r_{23}} = -\frac{k}{\sqrt{2}} + k\,.$$

Since $q_2 = 1\,\mathrm{C}$, the potential energy stored in q_2 is

$$U_2 = q_2 V = k\left(1 - \frac{1}{\sqrt{2}}\right) = 2.63 \cdot 10^9\,\mathrm{J}\,,$$

where I used $k = 8.99 \cdot 10^9\,\mathrm{Nm}^2/\mathrm{C}^2$ in the last step.

Part (e) To assemble the three-charge system, we should add the energy needed in each step, without double-counting the same pair of charges. We do this following the hint in the problem.

Step 1: Starting from empty space, bring in q_1. Since there are no other charges yet, it is really really really easy to move q_1 around in space, this costs

$$W_1 = 0\,\mathrm{J}\,.$$

Step 2: Now bring in q_2 while q_1 is already fixed at the origin. Then q_2 will feel an electric force due to q_1, and that affects how much energy we need to put in when trying to move q_2 from infinity to the target location at $\vec{x}_2$. As we have seen above (similar to the calculation that we have for bringing in q_3), at infinity, q_2 has zero potential energy, thus we only care about the potential energy of q_2 at the target location $\vec{x}_2 = (1, -1)$. At the target location, the distance between q_1 and q_2 is $\sqrt{2}\,\mathrm{m}$, so

$$W_2 = U_f - U_i = U_f - 0 = U_f = \frac{kq_1q_2}{\sqrt{2}} = -\frac{k}{\sqrt{2}}\,.$$

This is negative because q_1 and q_2 have opposite signs, so the attractive electric force between q_1 and q_2 helps us bring q_2 in.

Step 3: Now bring in q_3 while q_1 and q_2 are already fixed. We have calculated the energy needed to bring in q_3 in part (c), this is

$$W_3 = 0\,\mathrm{J}\,.$$

Therefore, the total energy needed to assemble the system is

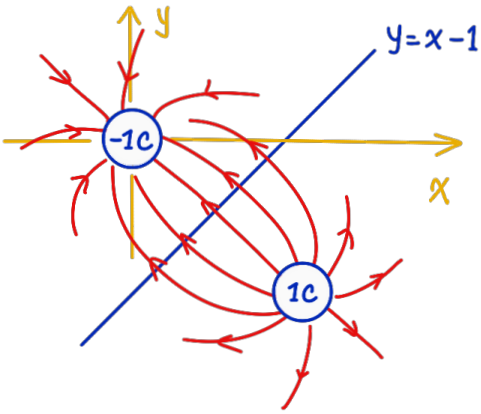
$$W_{\text{total}} = W_1 + W_2 + W_3 = -\frac{k}{\sqrt{2}} = -6.36 \cdot 10^9\,\mathrm{J}\,.$$

Equivalently, we can add the potential energies of the three pairs:

$$U_{\text{total}} = \frac{kq_1q_2}{r_{12}} + \frac{kq_1q_3}{r_{13}} + \frac{kq_2q_3}{r_{23}} = -\frac{k}{\sqrt{2}} - k + k = -\frac{k}{\sqrt{2}} = -6.36 \cdot 10^9\,\text{J}\,.$$

This gives the same result.

Part (f) Qualitatively, the electric field lines should start from the positive charge q_2 and end on the negative charge q_1. Since q_1 and q_2 have equal magnitude, every field line that leaves q_2 should eventually terminate on q_1. The picture should look like a dipole field, tilted along the line connecting $(1, -1)$ and $(0, 0)$.



Part (g) At $\vec{x}_i = (\infty, 0)\,\text{m}$, infinitely far away from the two source charges, the "distances" in

$$\vec{E} = \frac{kq_1}{(\text{“distance”})^2} + \frac{kq_2}{(\text{“distance”})^2}$$

are really large such that the fractions are essentially zero. Therefore, the electric field is

$$\vec{E}_i = (0, 0)\,\text{N/C}\,.$$

At $\vec{x}_f = (1, 0)\,\text{m}$, we compute the electric field due to each source charge. Considering q_1 only, the negative charge q_1 attracts a positive test charge toward the origin, so the field at $\vec{x}_f$ due to q_1 points directly left,

$$\vec{E}_1 = \left(-\frac{k|q_1|}{r^2}\,,\ 0\right) = (-8.99 \cdot 10^9\,,\ 0)\,\text{N/C}\,.$$

The positive charge q_2 pushes a positive test charge away from $(1, -1)$, so at $\vec{x}_f = (1, 0)$ the field due to q_2 points directly upward,

$$\vec{E}_2 = \left(0\,,\ \frac{k|q_2|}{r^2}\right) = (0\,,\ 8.99 \cdot 10^9)\,\text{N/C}\,.$$

Adding the two contributions,

$$\vec{E}_f = \vec{E}_1 + \vec{E}_2 = (-8.99 \cdot 10^9,\ 8.99 \cdot 10^9)\,\text{N/C}\,.$$

So the electric field at $\vec{x}_f$ is not zero, even though the electric potential at $\vec{x}_f$ is zero.

Part (h) The electric force on q_3 is

$$\vec{F} = q_3 \vec{E}\,.$$

We only need to scale what we have obtained for $\vec{E}$ at the initial and final positions. Since $q_3 = 1\,\mathrm{C}$,

$$\vec{F}_i = (0,0)\,\mathrm{N}\,, \qquad \vec{F}_f = (-8.99 \cdot 10^9,\ 8.99 \cdot 10^9)\,\mathrm{N}\,.$$

The force at $\vec{x}_f$ points up and to the left.

Part (i) If Jinyan flips the sign of q_3 at $\vec{x}_f$, then **the electric potential V and the electric field $\vec{E}$ due to the source charges q_1 and q_2 do not change**. These quantities are properties of the source charges, so they are defined by how and where you put the source charges. They have no relation with the test charge q_3.

What does change is the force on q_3, because

$$\vec{F} = q_3 \vec{E}\,.$$

After the sign flip, $q_3 = -1\,\mathrm{C}$, so at $\vec{x}_f$,

$$\vec{F}_f = (-1) \cdot (-8.99 \cdot 10^9,\ 8.99 \cdot 10^9)\,\mathrm{N}\,.$$

Thus the force reverses direction (the field does not reverse direction, and the potential does not change sign). The potential energy $U = q_3 V$ also changes sign in general, but at $\vec{x}_f$ we have $V_f = 0$, so in this special location the potential energy is still zero.

Part (j) The net electric potential due to q_1 and q_2 is zero whenever the two contributions from q_1 and q_2 cancel,

$$\frac{kq_1}{r_1} + \frac{kq_2}{r_2} = 0\,.$$

Since $q_1 = -1\,\mathrm{C}$ and $q_2 = 1\,\mathrm{C}$, this becomes

$$-\frac{k}{r_1} + \frac{k}{r_2} = 0\,,$$

so we need

$$r_1 = r_2\,.$$

In words, the net potential is zero at every point that is equally far away from q_1 and q_2. Geometrically, if you plot some of these points, you see that the line

$$y = x - 1$$

in the coordinate system captures all of them. Therefore, there is a whole line where the potential is zero. One convenient point on this line is the point we already used, $\vec{x}_f = (1,0)\,\mathrm{m}$. At that point, from part (g),

$$\vec{E}_f = (-8.99 \cdot 10^9,\ 8.99 \cdot 10^9)\,\mathrm{N/C}\,,$$

so the magnitude of the electric field is

$$|\vec{E}_f| = \sqrt{(-8.99 \cdot 10^9)^2 + (8.99 \cdot 10^9)^2} = 1.27 \cdot 10^{10}\,\mathrm{N/C}\,.$$

The electric potential being zero does not imply electric field being zero. Rather, at a specific location, electric field tells us which way and how much the potential changes nearby (mathematically, this is called the gradient). As we discussed above, a close analogy to V is a mountain (or a sink for V produced by negative charge). A solid sphere in the gravity picture can stand at zero height but still be on a steep hill rolling towards negative height. In the electric case, the electric field points from higher potential toward lower potential, and it is perpendicular to the equipotential lines.

Mathematically, [2]

$$\vec{E} = -\nabla V = \left(-\frac{\partial V}{\partial x}\,,\ -\frac{\partial V}{\partial y}\right) \approx \left(-\frac{\Delta V}{\Delta x}\,,\ -\frac{\Delta V}{\Delta y}\right)$$

As you can see here, $\vec{E}$ describes how V changes in all directions, so only the "change" matters. Two different lines $f(x) = 114x$ and $f(x) = 114x + 514$ have the same slope of 114 at every single point, which describes the change of $f(x)$ with respect to x, but at different points they satisfy $f(x) = 0$. The same argument applies to the relation between $\vec{E}$ and V.

[2] Review the week-4 worksheet for practical computation of this quantity.